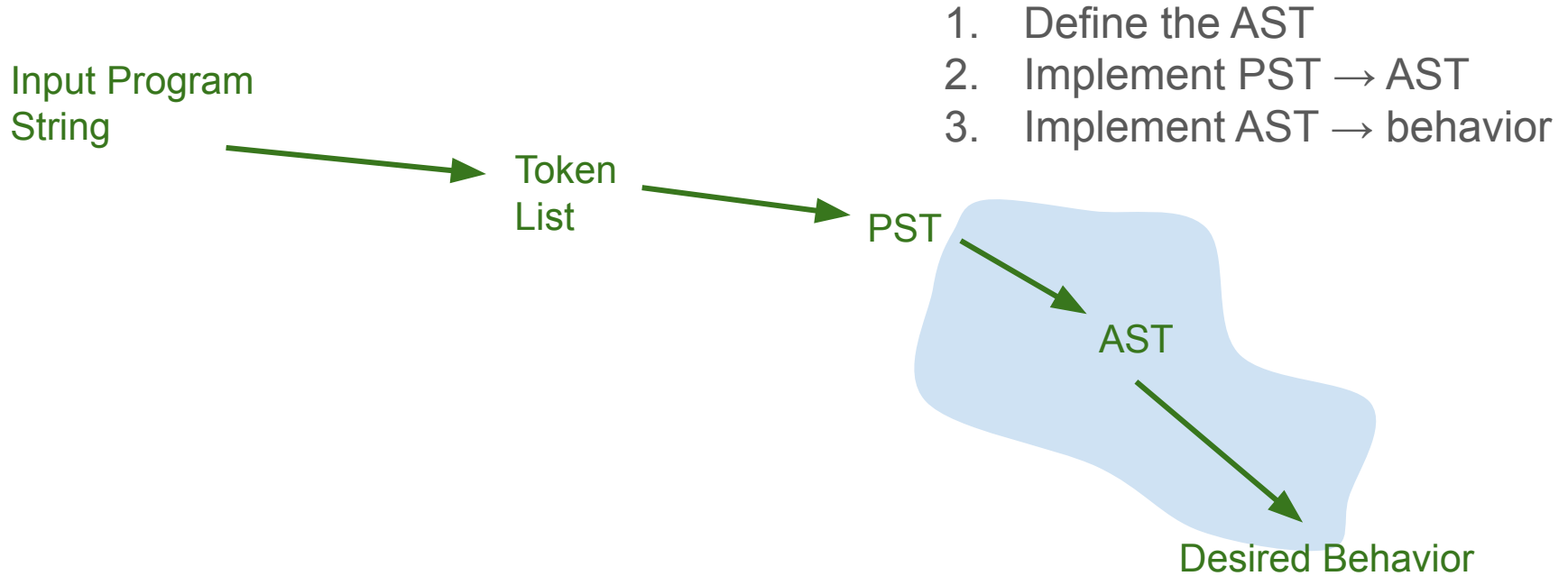


Lecture 16

Trefoil v2 Expressions

How to Add a Language Feature



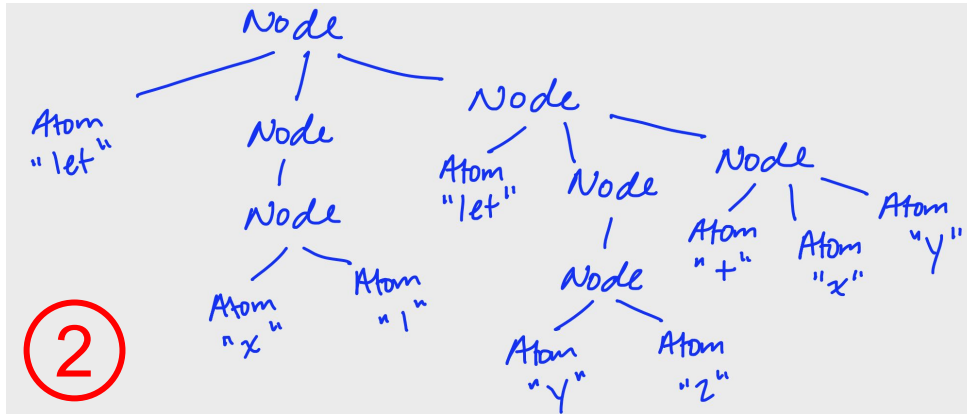
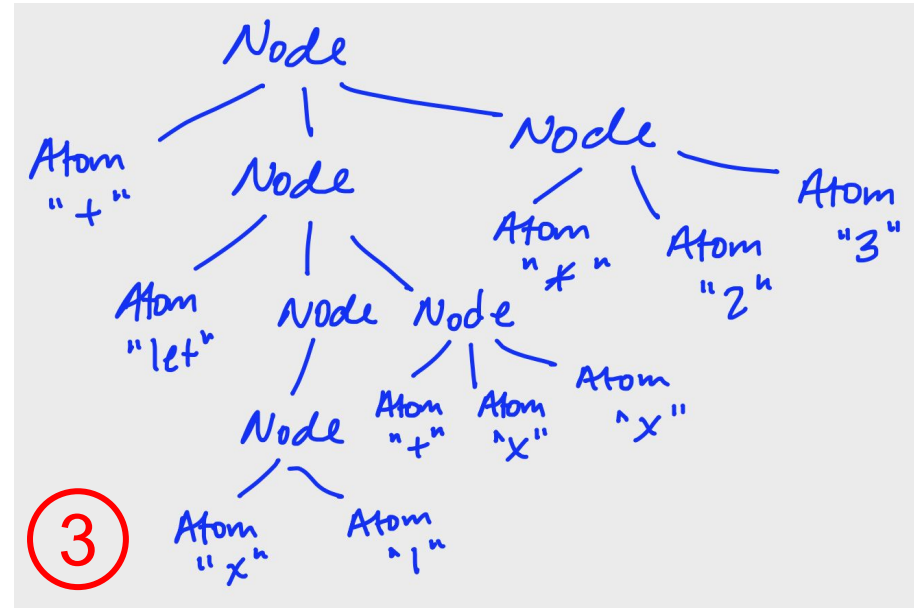
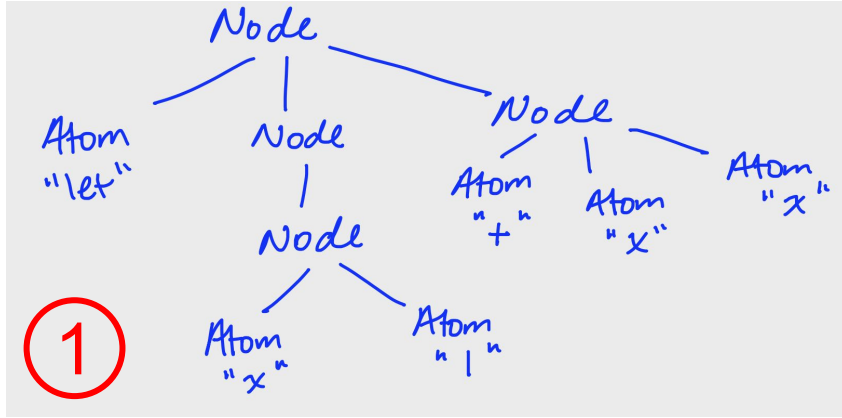
Group Work

```
type pst =  
  | Atom of string  
  | Node of pst list
```

Draw the PSTs for the following Trefoil expressions:

1. `(let ((x 1)) (+ x x))`
2. `(let ((x 1)) (let ((y 2)) (+ x y)))`
3. `(+ (let ((x 1)) (+ x x)) (* 2 3))`

Group Work 1 Answers



Group Work

	1	2	3
A			
B			
C			

Here are 3 possible implementations of Let expressions:

1. `| Let (x, e1, e2) -> let v = interp env e1 in interp env e2`
2. `| Let (x, e1, e2) -> let v = interp empty e1 in interp empty e2`
3. `| Let (x, e1, e2) ->
 let v = interp env e1 in interp (bind x v empty) e2`

Here are 3 example ASTs:

- A. `Let ("x", Int 1, Add (Var "x", Int 4))`
- B. `Let ("x", Int 1, Let ("y", Int 2, Add (Var "x", Var "y")))`
- C. `Let("x", Int 1, Add(
 Let ("x", Int 2, Add (Var "x", Var "x")),
 Mul (Var "x", Int 5)))`

What happens when you interpret each AST with each implementation?

Group Work 2 Answers

	1	2	3
A	error	error	error
B	error	error	error
C	5	error	9